KINGDOM OF SAUDI ARABIA
Ministry of Education
King Khalid University
College of Computer Science

المملكة العربية السعودية
وزارة التعليم
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Facial Emotion Recognition using Machine Learning

By

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Submitted to King Khalid University in fulfilment of the requirements for the B.Sc. Degree in Information Systems

Academic Year 2023-24/1444-1445 (Semester I)
Department of Information Systems
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Abstract

Facial Emotion Recognition Using Machine Learning is a research project focused on developing a reliable system for detecting and recognizing emotions from facial expressions.[12] The goal is to leverage machine learning techniques to provide an accurate and effective human emotion reading solution to understand customers' emotions while shopping. [13] Every e-commerce company wants their customers to come back and buy again, if not now, then in a month. Every purchase is an opportunity to learn what customers are looking for, what they might want, and why they made specific purchasing decisions.[14]

This research addresses the need for a reliable method to interpret and understand human emotions based on facial expressions using machine learning and artificial intelligence.

The motivation behind this research stems from the desire to improve sales and customer satisfaction and knowing which products are more welcomed by customers and helps in product classification.

In taking advantage of advances in machine learning to enhance our understanding of human emotions and improve various applications.[15] Through accurate recognition of facial expressions.

The approach involves scanning and analysing existing facial expression recognition technologies to gain insight into their effectiveness.[16] Based on the results, a method based on machine learning or facial artificial intelligence for expression recognition will be implemented.[17] The chosen technology will be trained using labelled datasets, enabling it to accurately identify and classify different emotions displayed on the face in images.[10]

Evaluation of the developed method involves testing its performance on different datasets and scenarios.[18] The system's ability to accurately detect and interpret emotions will be evaluated using appropriate evaluation metrics. The project also realizes the potential benefits of increasing sales.[19]

Moreover, this project recognizes the importance of understanding human emotions and aims to facilitate product classification and understand customer emotions through accurate recognition of facial expressions and better interpretation of people's emotions.[20]

In conclusion, facial emotion recognition using machine learning is a research project that aims to develop a reliable system for detecting and recognizing emotions through facial expressions.[21]

Declarations

We are here to declare that the project entitled “Facial Emotion Recognition using Machine Learning “is based on our original work carried out during our study under the supervision of Dr. SUHAIL QADIR GH QADIR. We have not copied from any other sources except where due references or acknowledgment are made explicitly in the text, nor has any part been written for me by another person.

Acknowledgements

Praise be to God, we completed the first part of the project with a wonderful and cooperative team in the required time. We thank the project supervisor Dr. SUHAIL QADIR GH QADIR , for his continuous support and guidance, We thank everyone who was the reason for the success of this project.

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Chapter 1: Introduction and Background

1.1 Goal

The goal of this project is to design an application model capable of accurately predicting and understanding customer emotions across seven categories: anger, disgust, fear, happiness, neutral, sadness, and surprise.

1.2 Motivation

The motivation behind this research stems from a desire to take advantage of advances in machine learning to enhance our understanding of customer emotions while shopping and improve various applications. Through accurate recognition of facial expressions.

1.3 Significance

The importance of this project lies in its ability to understand human feelings. It will help us understand customer feelings and will help companies with customer satisfaction and in increasing sales and other matters and responding to them through visual observation. Additionally, it can have practical implications and these studies have shown promising results in detecting emotions such as anger, fear, joy, and sadness from facial expressions.

1.4 Method

Data collection and preparation: The first step will be to collect a dataset of facial images with corresponding emotion labels. This dataset can be collected from public sources or by manually annotating images.

Feature extraction: The next step is to extract features from the facial images. This can be done using handcrafted features or machine learning techniques.

Model training: The extracted features will then be used to train a machine learning model. This model can be a support vector machine, a neural network, or another type of machine learning algorithm.

Model evaluation: The trained model will then be evaluated on a test set of images. This will help to determine the accuracy of the model.

Deployment: The final step is to deploy the model in a real-world application. This could involve using the model to control a robot or to provide feedback to a user.

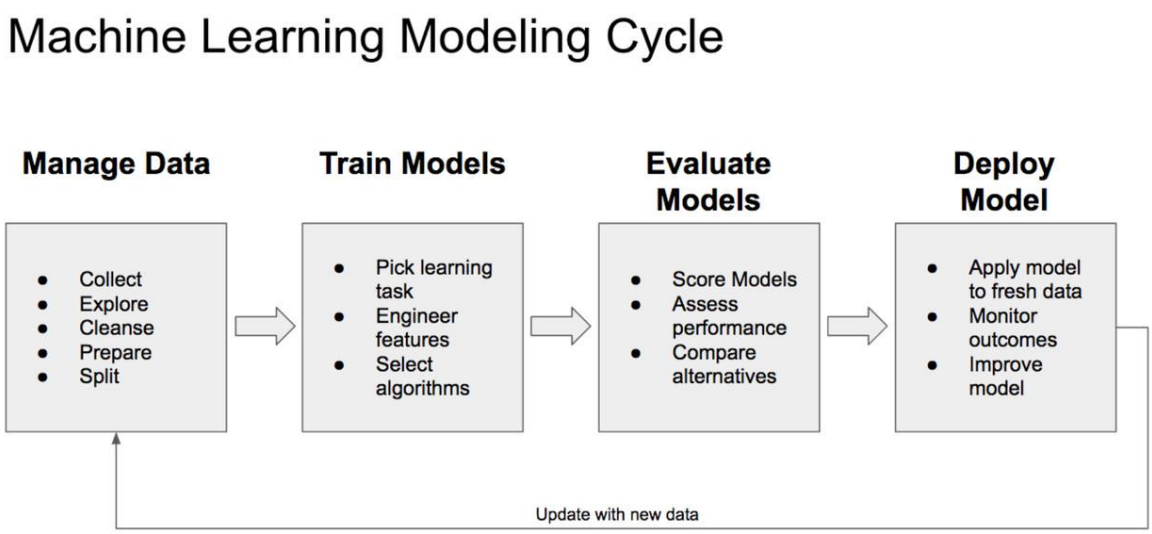

Figure 1 method for ML

- **Gantt chart:**

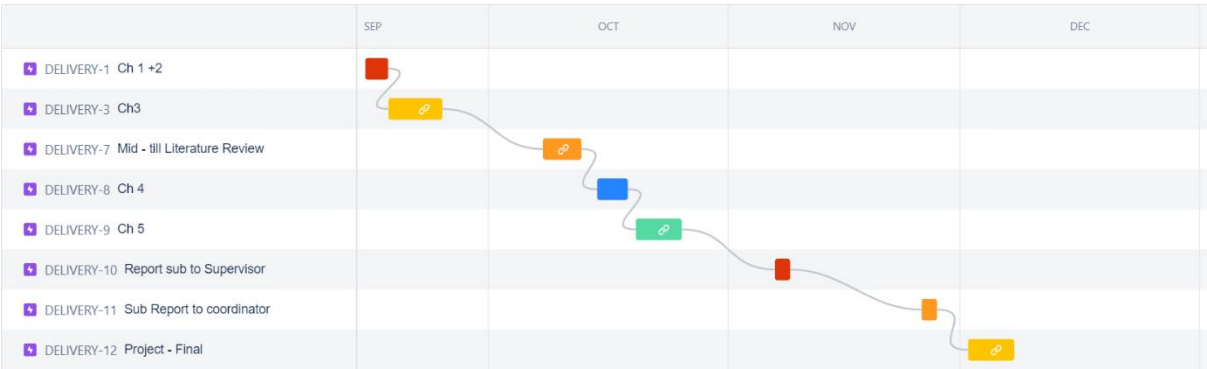

Figure 2 Gannt Chart

- **Work Breakdown Structure**

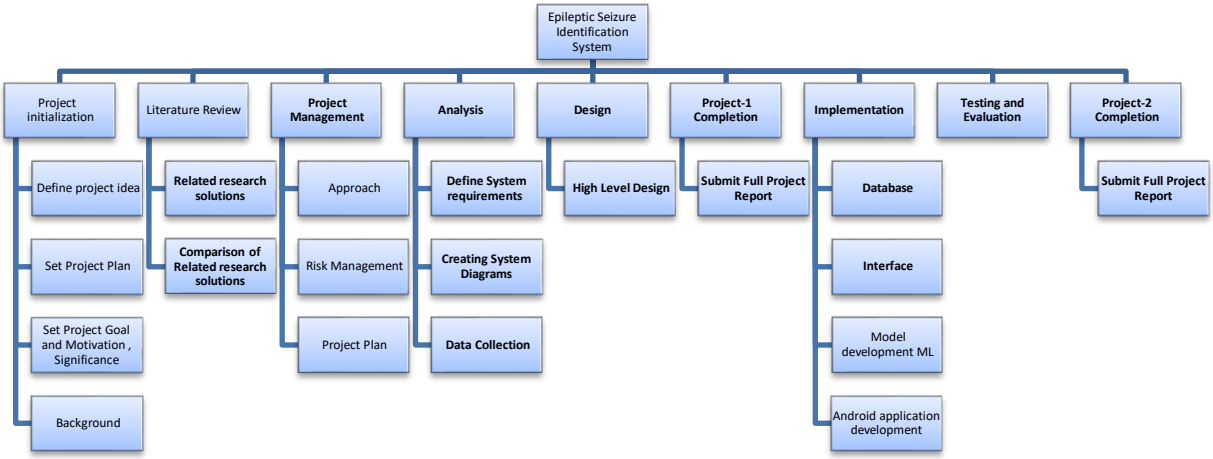

Figure 3 : Work Breakdown Structure

1.5 Problem Statement

Facial emotion recognition is the task of identifying and classifying emotions through facial expressions, feature analysis, and facial geometry [11]. It is a difficult task due to the complexity and diversity of facial expressions. We will need to collect data and images to create a good model, and we will need to try to obtain the highest accuracy.

1.6 Background Information

Machine learning techniques provide a significant improvement in emotion prediction through recognition of facial expressions,[10] physiological signals, and other indicators. Models are trained on massive datasets to learn patterns and features that enable accurate prediction of sentiment in live scenarios.

1.6.0 Artificial Intelligence

Artificial Intelligence (AI) refers to the development of intelligent systems that can perform tasks requiring human-like intelligence.[7] It involves the creation of algorithms and models that enable machines to understand, reason, learn, and make decisions. In the context of facial emotion recognition, AI techniques are used to develop systems that can analyze facial

expressions [8] and interpret the corresponding emotions. AI algorithms can be trained on large datasets to learn patterns and relationships between facial features and emotions, allowing for automated recognition and interpretation.

1.6.1 Machine Learning

Machine Learning (ML) is a subset of AI that focuses on developing algorithms and models that enable computers to learn and make predictions or decisions without being explicitly programmed. ML algorithms can learn from data and improve their performance over time through experience. In the context of facial emotion recognition, ML techniques are used to train models on labelled datasets of facial images. These models can then generalize from the learned patterns to recognize and classify emotions in new images or videos. ML algorithms commonly used in facial emotion recognition include deep learning algorithms such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs).[2][3]

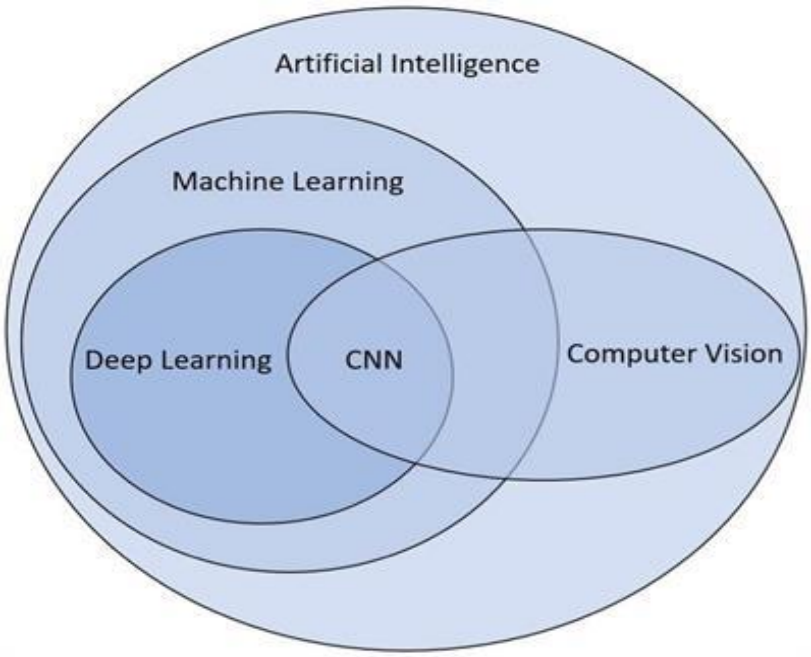

Figure 4 Background Information

1.7 Future of Facial Emotion Recognition

The field of facial emotion recognition is rapidly evolving. Recent advances in machine learning have led to significant improvements in the accuracy of facial emotion recognition systems. As the technology continues to develop, [9] it is likely that facial emotion recognition will become more widespread and used in a wider range of applications.

1.8 Report Outline

Briefly outline the remainder of this report in a paragraph.

This thesis is organised as follows.

Chapter 2: Literature Review - This chapter provides an overview of theories, concepts, and research relevant to the project.

Chapter 3: Project Management - This chapter discusses the project management approach, including planning, scheduling, and resource allocation.

Chapter 0: Requirements and Analysis - This chapter focuses on gathering and analysing requirements for an emotion detection model, considering factors such as accuracy, efficiency, and ease of use.

Chapter 5: System Design - This chapter presents the detailed design of an emotion detection system, including Data Flow Diagram, UML Use Case Diagram, UML Class Diagram, Activity Diagram.

Chapter 2: LITERATURE REVIEW

2.0 Introduction

Facial emotion recognition (FER) is a research field that aims to develop automated systems that can accurately recognize human emotions through facial expressions [1]. FER has a wide range of potential applications.

machine learning (ML) techniques have proven to be very effective in FER tasks [1]. A variety of machine learning models have been applied in FER, including:

1. **Convolutional Neural Networks (CNNs):** CNNs have been widely used for FER due to their ability to capture spatial dependencies in facial images. They can automatically learn discriminative features from raw pixel values and have achieved state-of-the-art performance in FER tasks [2].

2. **Recurrent Neural Networks (RNNs**): RNNs, particularly Long Short-Term Memory (LSTM) networks, have been employed to capture temporal dependencies in facial expression sequences. They can model the dynamics of facial expressions over time and have shown promising results in FER [3].

3. **Support Vector Machines (SVMs):** SVMs have been utilized for FER by mapping facial expression features to a higher-dimensional space and finding an optimal hyperplane for classification. SVMs have demonstrated good generalization performance and have been widely used in FER research [4].

4. **Random Forests**: Random Forests are ensemble learning models that combine multiple decision trees to make predictions. They have been applied in FER to handle high-dimensional feature spaces and have shown robust performance in emotion recognition [5].

5. **Deep Belief Networks (DBNs):** DBNs are deep learning models that consist of multiple layers of hidden units. They have been employed in FER to learn hierarchical representations of facial expressions and have achieved competitive results in emotion recognition tasks [6].

This chapter will review the state-of-the-art in FER using machine learning techniques. We will discuss the different types of ML algorithms that have been used for FER, the different types of facial features that are typically extracted, and the evaluation metrics used to evaluate the performance of FER systems.

2.1 Related research solutions

2.1.0" Emotion AI analysing user while watching video." Website

This solution focuses on analysing user emotions while they are watching a video. It utilizes Emotion AI techniques to detect and interpret emotions exhibited by the user. The system provides a dashboard that presents the analysis of emotions in a visual format, allowing for easy interpretation and understanding of the emotional responses. Figure 3 illustrates the Emotion AI system in action, analysing the user's emotions while watching a video. Figure 4 showcases the dashboard that provides a comprehensive view of the emotion analysis results.

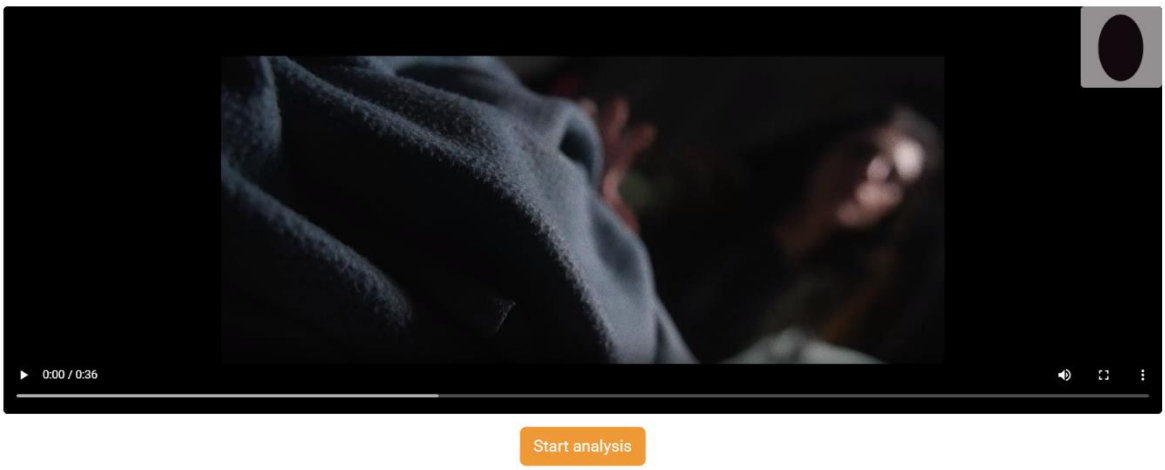

Figure 5 Emotion AI analysing user while watching video

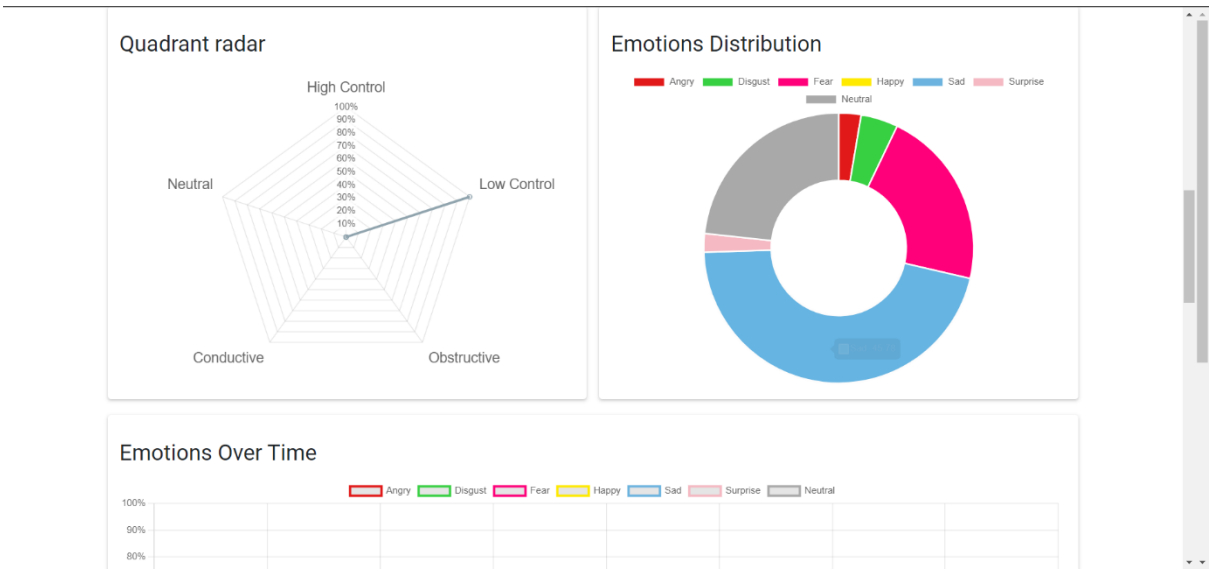

Figure 6 Emotion AI analysing user while watching video – Dashbord

2.1.1" Facial Emotion Recognition AI" website

This research solution centres around facial emotion recognition using artificial intelligence. The system is designed to analyse facial expressions in real-life scenarios, capturing emotions displayed by individuals. It employs advanced AI algorithms and techniques to accurately recognize and classify different emotions based on facial cues. Figure 5 represents the Facial Emotion Recognition AI system, which can provide real-time analysis of emotions from facial expressions.

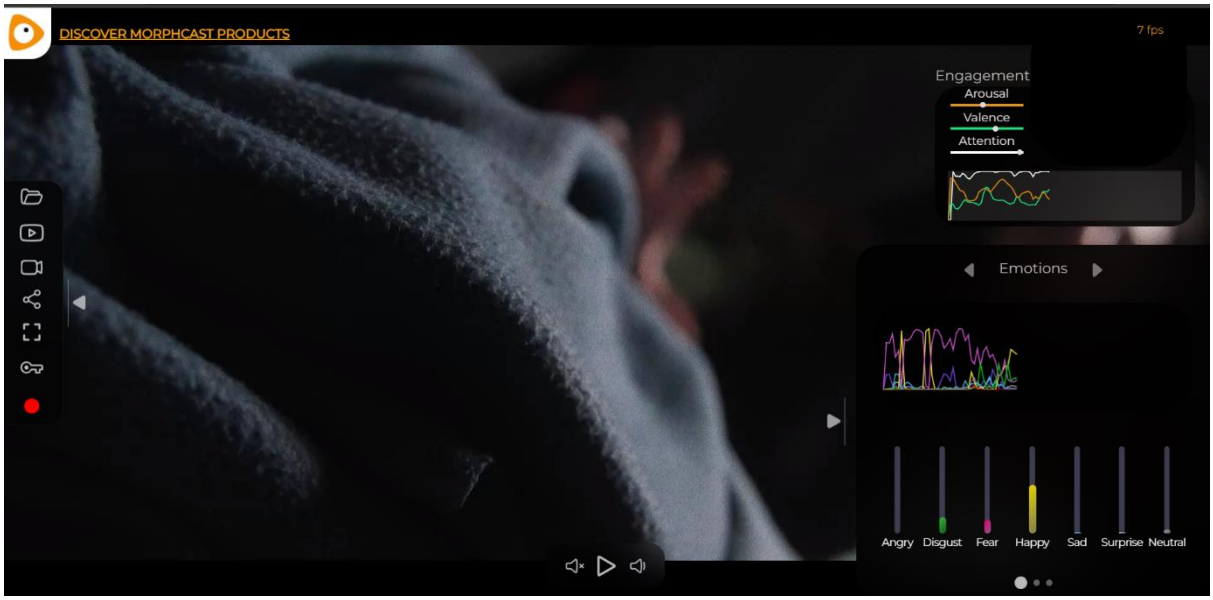

Figure 7 Facial Emotion Recognition AI

2.1.2 "Expression AI Emotion Finder" Android application

AI powered real time facial emotion analyzer to detect emotions in photos and camera feeds, all on device.

Here is where the fun begins.

Find up to 8 different emotions and 45+ different sentiments along with attention and excitedness levels in any photo.

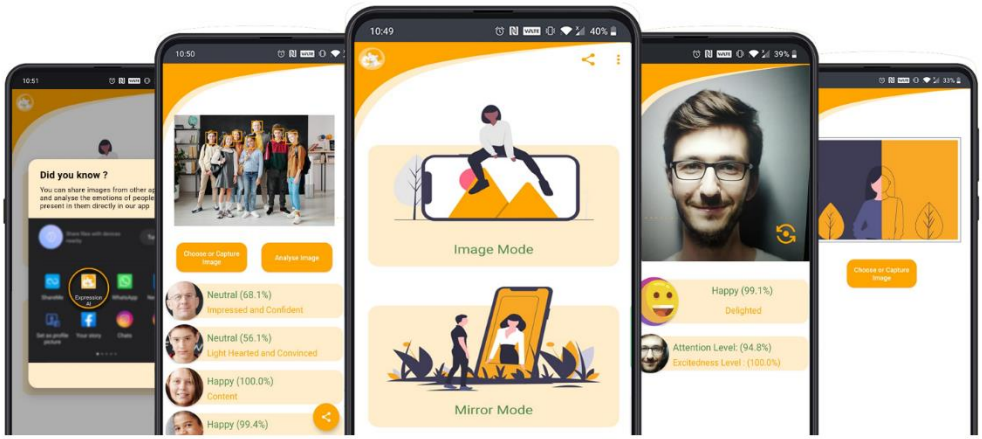

Figure 8 Expression AI Emotion Finder App

2.1.3 " Emotimeter - Emotion detector " Android application

Emotimeter can detect emotions from facial expressions using cutting edge machine learning technologies.

With Emotimeter you can:

- Detect emotions in real-time from the images obtained by the camera.

- Take photos or record videos of these live detection sessions.

- Analyze photos from your gallery to detect the emotions of all the people in it.

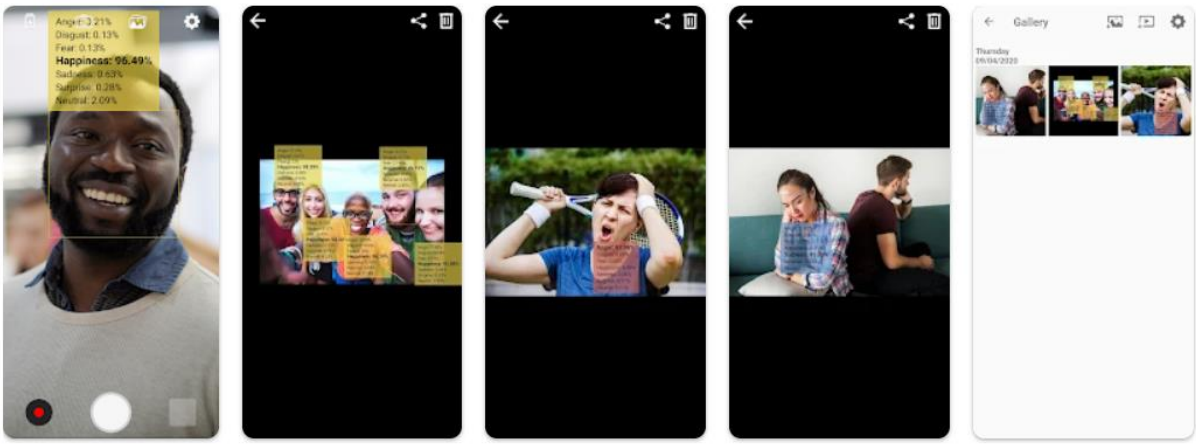

Figure 9 Emotimeter - Emotion detector " Android application

2.2 Comparison of Related research solutions

System / Feature	Expression AI Emotion Finder	Emotimeter	Facial Emotion Recognition AI	Emotion AI analyzing user while watching video	Proposed system
Is it easy to use and to, learn?	***No***	***Yes***	***Yes***	***Yes***	***Yes***
Does it support the Arabic language inall its features?	***No***	***No***	***No***	***No***	***No***
Are all features free?	***Yes***	***Yes***	***No***	***No***	***Yes***
Does it provide statistical analysis?	***Yes***	***Yes***	***Yes***	***Yes***	***Yes***
Does it provide the dashboard?	***No***	***No***	***Yes***	***Yes***	***Yes***
Does it provide high accuracy?	***Yes***	***Yes***	***Yes***	***Yes***	***Yes***
Can the user get support easily?	***No***	***No***	***Yes***	***No***	***Yes***

Table 1 Comparison of Related research solutions

2.3 Added Value

- Ease of understanding customers Feling
- Improve customer service.
- easy to use.
- All features are free.

2.4 Research Questions

1. Which machine learning method will be useful for recognition emotional?
2. How to apply the machine learning techniques for object recognition?
3. How to evaluate the performance of the proposed system?
4. What are the main machine learning techniques used in Facial Emotion Recognition (FER)?

2.5 Resources and References:

[1] *(Machine learning based customer sentiment analysis for recommending shoppers, shops based on customers' review)*

https://link.springer.com/article/10.1007/s40747-020-00155-2

[2] Reference for CNN-based FER model.

[3] Reference for RNN-based FER model.

[4] Reference for SVM-based FER model.

[5] Reference for Random Forest-based FER model.

[6] Reference for DBN-based FER model

[7] https://www.frontiersin.org/articles/10.3389/frai.2021.622364/full

[8] https://ieeexplore.ieee.org/abstract/document/9091188

[9] https://link.springer.com/chapter/10.1007/978-3-031-05409-9_25

[10] https://mdpi.com/1999-5903/11/5/105

[11] https://www.sciencedirect.com/science/article/abs/pii/S0031320313004202

[12] https://www.mdpi.com/1424-8220/18/2/416

[13] https://link.springer.com/article/10.1007/s40747-020-00155-2

[14] https://link.springer.com/article/10.1177/009207002236918

[15] https://ieeexplore.ieee.org/document/8849996

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[17] https://www.mdpi.com/2076-3417/11/19/9174

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[19]https://workplaceif.com/wp-content/uploads/2017/09/HBR_Competing_on_Talent_Analytics.pdf

[20] https://journals.sagepub.com/doi/abs/10.1509/jmkg.75.3.132

[21] https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0247131

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[22] https://ietresearch.onlinelibrary.wiley.com/doi/full/10.1049/iet-ipr.2018.6647

[23] https://www.sciencedirect.com/science/article/abs/pii/S0019850116303376

https://www.theatlantic.com/technology/archive/2014/01/this-app-reads-your-emotions-on-your-face/282993/

https://www.mdpi.com/2073-431X/12/2/37

https://www.kdnuggets.com/2022/07/bounding-box-deep-learning-future-video-annotation.html

Chapter 3: PROJECT MANAGEMENT

3.0 Approach

We have adopted the Agile methodology as our approach to project management. Agile methodologies prioritize flexibility, collaboration, and iterative development.[21] This approach allows us to adapt to changing requirements and deliver incremental value to stakeholders throughout the project lifecycle. By employing Agile, we can respond to feedback effectively and ensure that the project stays aligned with the evolving needs of the stakeholders.

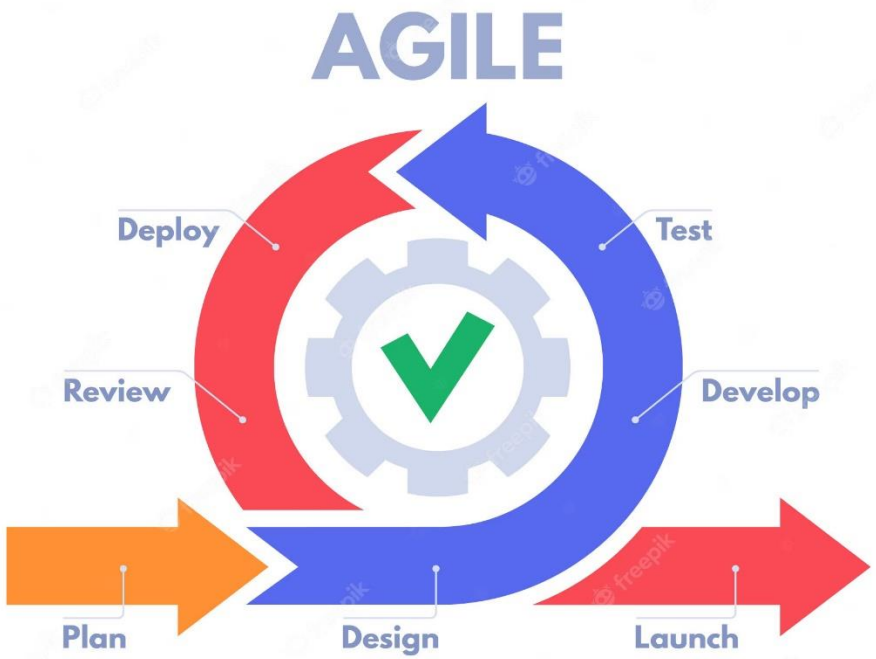

Figure 10 Agile methodologies

3.0.0 Agile methodology

Agile Software Development lifecycle phases:

Requirements: Stakeholders in this phase will determine the time and resources required for the development process. Also, will assesses the risks and prioritizes the various functions depending on their business value.

Design: The stakeholder meets with the software development team and presents them to the requirements outlined in the first step. then the group discusses the sequence for introducing functions and identifies the essential tools. At the same stage, software development teams can prototype the expected user interface.

Development: After agreeing on the plan, the team develops the product. The product is delivered in stages, in separate sprints, each designed to improve the current version of the product.

Testing: At this phase, the product becomes available to consumers and their feedback will collect, to ensure that the software is fully functional, the team must conduct a series of tests. If potential bugs or flaws are found, the developers will fix them immediately.

Deployment: The software is now fully deployed and available to customers. During this phase, the software development team provides ongoing support to keep the system running smoothly and fix any new bugs.

Review: After completing all the previous stages of development, the result achieved in meeting the requirements will be presented by the development team to the owner. Then the Agile phases start over – either with a new iteration or moving to the next stage and scaling Agile.

3.1 Risk Management

- Data Quality
- Model Overfitting
- Lack of Representative Training Data
- Algorithm Selection
- Ethical and Privacy Concerns

3.2 Project Plan

Task Name	Member
Initializing project	Rimas ,Manar,Ghadi
Defining project idea	
Setting project goal ,motivation	
Background	
Literature Review	Rimas ,Manar,Ghadi
Related research solution	
Comparison of related research	
Project Management	Rimas ,Manar,Ghadi
Approach	
Risk Management	
Project Plan	
Analysis	Rimas ,Manar,Ghadi
Define system requirements	
Creating system Diagrams	
Data collection	
Design	Rimas ,Manar,Ghadi
High Level Design	
Project-1 completion	Rimas ,Manar,Ghadi
Submit report	Rimas
Implementation	Rimas ,Manar,Ghadi
Database	
Interface	
Model development ML	
Android application development	
Testing and Evaluation	Rimas ,Manar,Ghadi
Project-2 completion	Rimas ,Manar,Ghadi
Submit report	Rimas

Table 2 Project Plan and task

Chapter 4: Requirements and Analysis

4.0.0 Introduction

Facial expression recognition (FER) is a technology that can be used to identify and analyse facial expressions in real time.[22] It has the potential to be used in a variety of customer-facing applications, such as customer service, product development, and marketing and sales.[23]

4.1.0 Requirement Elicitation

Component	Description
Task	Analyse customers' facial expressions to assess their satisfaction with services.
Input	Customers' faces
Output	Summary of customers' facial expressions with 7 categories
Benefits	Real-time feedback, unobtrusive assessment, broader range of emotions
Challenges	Accuracy, privacy concerns, interpretation

Table 3 Requirement Elicitation

4.1.1 Quantitative and Qualitative Data Analysis

Objective	Quantitative Analysis	Qualitative Analysis	Objective
Understand user needs	Collect data on behaviours and interactions	Collect data on feedback sentiment	Understand user needs
Improve the product or service	Identify strengths and weaknesses	Understand the underlying causes of behaviours	Improve the product or service

Table 4 Quantitative and Qualitative Data Analysis

4.1.2 Problems and Solutions:

Identifier	Name	Problem	Solution	Priority
FER - 001	Capturing customer emotions with a camera	The camera needs to capture client emotions accurately and effectively.	Use facial recognition and emotional analysis techniques to analyse facial expressions and emotions.	High
FER - 002	Send sentiments to the system for analysis	Camera data for sentiment analysis needs to be sent to the system.	Establish communication between the camera and the system via an API for data	High

			transmission and analysis.	
FER - 003	Display emotions on the dashboard	Analysed sentiment needs to be displayed on a user-friendly and real-time dashboard.	Develop a dashboard that presents the analysed sentiment in an easy-to-understand manner and allows customers to view it in real time.	High
FER-005	Enhance and refine the labelled dataset	Enhance the quality and consistency of the labelled dataset to improve the performance of facial expression recognition (FER) models.	Step 1: Review and refine emotion categories. Step 2: Update annotation guidelines. Step 3: Assess data quality. Step 4: Enhance data diversity.	High
FER - 006	Train models on high-performance computing platforms	Training deep learning models for FER requires significant computational resources.	Use high-capacity computers and access high-performance computing platforms.	High
FER - 007	Integrate with existing customer-facing applications	Integrating FER systems with existing customer-facing applications can be challenging.	Address system compatibility and ensure seamless integration.	High
FER - 009	Optimize FER systems for real-time performance	FER systems need to be able to process video streams and detect facial expressions in real time.	Optimize FER systems for real-time performance using techniques such as parallel processing and model compression.	High

Table 5 Problems and Solutions

4.2.0 System Requirements

System requirements for a facial emotion recognition (FER) system can include:

Requirement	Description
Image Upload	Users must upload their images to the application. The images must be in JPG or JPEG.
Emotion Recognition	The application must recognize emotions from the uploaded images. The application must support the recognition of the following emotions: happiness, sadness, anger, fear, surprise, and disgust.
Results Display	The results of the emotion recognition must be displayed on the dashboard.
Real-time facial expression analysis	The system must continuously analyse facial expressions in real time

Table 6 System Requirements

4.2.1 Functional requirements

describe the specific functionalities and capabilities that a system or software must possess to meet the intended goals and objectives.

Requirement Type	Requirement	Description
Functional	FR1	The application must allow users to upload images to the application.
Functional	FR2	The application must recognize emotions from the uploaded images.
Functional	FR3	The application must support the recognition of the following emotions: happiness, sadness, anger, fear, surprise, and disgust.
Functional	FR4	The application must display the results of the emotion recognition on a dashboard.
Functional	FR5	The results of the emotion recognition must image and detected emotions.
Functional	FR6	The system must continuously analyse facial expressions in real-time, providing immediate feedback to customer service representatives.

Table 7 Functional requirements

4.2.2 The non-functional requirements

describe the qualities, characteristics, constraints, and criteria that a system should possess or adhere to.

Requirement Type	Requirement	Description
Non-Functional	NFR1	The application should be easy to use.
Non-Functional	NFR2	The application should be efficient.
Non-Functional	NFR3	The application should be accurate.
Non-Functional	NFR4	The application should be reliable.
Non-Functional	NFR5	The application should be secure.
Non-Functional	NFR6	The application should be scalable.

Table 8 The non-functional requirements

4.2.3 Hardware requirements

refer to the specific hardware components and specifications that are necessary for a system or software application to run effectively.

Requirement Type	Requirement	Description	Notes
Hardware	Processor	Intel Core i5-8400	Minimum requirement for basic functionality.
Hardware	Memory	8 GB	Minimum requirement for basic functionality.
Hardware	Storage	250 GB SSD	Minimum requirement for basic functionality.
Hardware	Graphics Card	NVIDIA GeForce GTX 1060	Recommended requirement for better performance and accuracy.
Hardware	Operating System	Windows 10 or later	Minimum requirement for basic functionality.

Table 9 Hardware requirements

4.2.4 Software requirements

refer to the specific functionalities, features, and qualities that a software application or system should possess to meet the desired objectives and user needs.

Requirement Type	Requirement	Description	Notes
Software	Software Development Environment	Flutter	Flutter is a modern cross-platform development framework that is well-suited for mobile applications.
Software	Programming Language	Dart	Dart is a programming language that is designed for web, mobile, and desktop applications.
Software	IDE for Android App Development	Android Studio	Android Studio is the official integrated development environment (IDE) for Android app development. It provides a comprehensive set of tools for building,

			testing, and debugging Android applications.
Software	Machine Learning Model	Python/TensorFlow	Python is a popular programming language for machine learning, and TensorFlow is a popular machine learning framework.
Software	IDE for Kaggle Model Development	Google Collaboratory (Kaggle Notebooks)	Google Collaboratory is a cloud-based Jupiter notebook environment that is well-suited for developing and running machine learning models. It provides access to powerful computing resources and a variety of pre-installed machine learning libraries, including TensorFlow.
Software	UI Toolkit (UI Development for Mobile App)	Flutter	Flutter is a cross-platform UI toolkit that is ideal for building native mobile applications. It uses the Dart programming language and provides a declarative way to build user interfaces.
Software	UI Toolkit (UI Development for Dashboard)	Flutter	Flutter is a powerful and versatile framework that can be used to create a wide variety of user interfaces, including dashboards. It offers cross-platform development, high performance, declarative UI, hot reload, a rich widget library, and Material Design support, making it a suitable choice for building dashboards.

Software	Database	Firebase Realtime Database	A cloud-based database that provides real-time data synchronization for mobile and web applications.

Table 10 Software requirements

4.3.0 Personas

who benefit from facial emotion recognition (FER) technology in the company can include:

Customer service team: The customer service team can utilize FER technology to understand the emotions of customers during interactions. They can analyse facial expressions and identify customer emotions such as happiness, anger, or sadness, to provide an enhanced customer experience and better meet their needs.

Decision-making team: The decision-making teams in the company can use FER technology to gain valuable insights into customer emotions and their interaction with products or services. They can use this data to make strategic decisions to improve products, develop marketing strategies, or enhance customer satisfaction.

Data analytics team: The data analytics team in the company can leverage FER technology to analyse customer emotion-related data. They can create detailed reports and analyses that highlight emotional trends and customer feedback, enabling the identification of strengths and weaknesses in the business process and facilitating appropriate actions.

Executive management: The executive management in the company can use FER technology to gain a comprehensive view of customer satisfaction and emotions towards products or services. They can monitor a dashboard that displays data extracted from FER technology, providing them with strategic insights to improve performance and enhance the customer experience.

Call centres: The system can be used to provide real-time feedback to call centre representatives about customer emotions.

4.4.0 System Models

To create a system model for a facial emotion recognition (FER) system, you can use either the traditional approach or the Object-Oriented (OO) approach. Here are the corresponding system models for each approach:

Traditional Approach	**OO Approach**
Data Flow Diagram *Create a DFD to depict the flow of data within the FER system. It will illustrate how data is processed and transferred between different components of the system.* .	**4.3.1 UML Use Case Diagram** Create a UML Use Case Diagram to define the system's boundaries and functionalities from the user's perspective. It will identify the actors interacting with the system and the use cases they perform.
Process Specification *Document the decision-making logic and formulas used to generate output data from input data. This may include using techniques like Structured English to describe each step of the process.*	**4.3.2 Process Specification** Describe the use case descriptions and scenarios in detail. You can also include UML sequence, state, and activity diagrams for important processes within the system.
Entity Relationship Diagram *Design an ERD to represent the entities and their relationships within the FER system. This diagram will show the structure of the system's data and how different entities are related.*	**4.3.3 UML Class Diagram** This diagram describes the static structure of the system in term of classes. All relationships among classes should be represented.

Chapter 5: System Design

This chapter describes the system design for a facial expression recognition (FER) system for customer satisfaction. The system will use machine learning algorithms to analyse facial expressions and identify customer emotions in real time. The system will be used by customer service representatives to better understand the needs of customers and provide better service.

5.0.0 Product Features

The facial expression recognition (FER) system for customer satisfaction will offer a range of features to enhance customer service interactions and improve customer satisfaction levels. These features include:

Real-time facial expression analysis: The system will continuously analyse facial expressions in real time, providing immediate feedback to customer service representatives.

Emotion detection: The system will identify and classify customer emotions, including happiness, sadness, anger, and surprise.

Emotion-based customer segmentation: The system will segment customers based on their emotional states, allowing customer service representatives to tailor their interactions accordingly.

Performance monitoring: The system will track the performance of customer service representatives, identifying areas for improvement.

5.1.0 User Interface

The FER system will feature an intuitive and user-friendly interface that is easy to navigate for customer service representatives. The interface will display the following key elements:

Live video feed: A live video feed of the customer's face will be displayed, allowing customer service representatives to observe facial expressions in real time.

Emotion indicators: Real-time emotion indicators will be displayed, providing immediate feedback on the customer's emotional state.

Image upload from device: A "Upload Image from Device" button will be added as an additional element in the intuitive and user-friendly interface. Customer service

representatives can upload images from their devices and use them to analyse and interpret emotions from images. The button will be displayed prominently and easily accessible in the interface, allowing representatives to easily upload images and examine them for emotion analysis and to better understand customer experience.

Performance dashboard: A performance dashboard will provide customer service representatives with insights into their performance and areas for improvement.

5.2.0 Data Storage

Data storage will include:

Facial image storage: Facial images will be stored temporarily for analysis and will be deleted promptly after processing.

Emotion data storage: Emotion data, including timestamps and emotion classifications, will be stored securely for further analysis and reporting.

Performance data storage: Customer service representative performance data will be stored to track progress and identify areas for improvement.

5.3.0 High Level Design

provides a view of the system at an abstract level.

5.3.1 UML Class Diagram

The UML class diagram provides a high-level overview of the classes and their relationships in the FER system. The diagram shows that the system is composed of several classes, including:

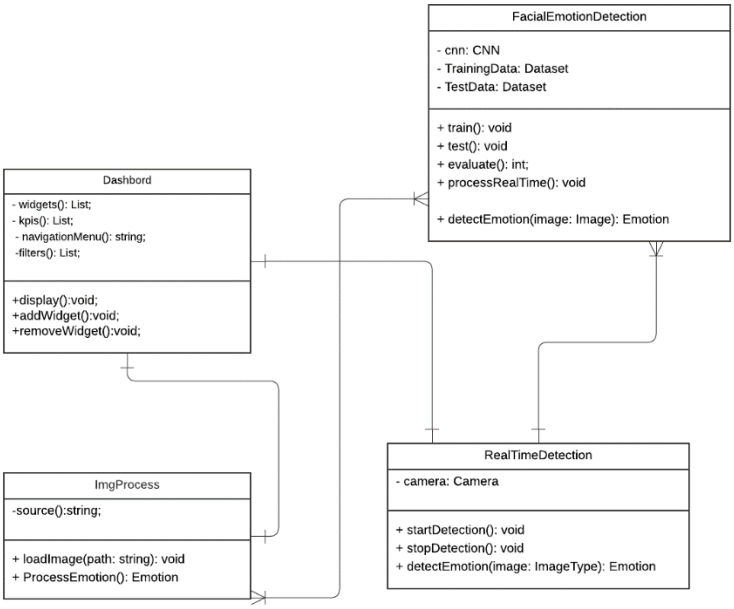

Figure 11 UML Class Diagram

5.3.2 Data Flow Diagram

The data flow diagram illustrates the flow of data through the FER system.

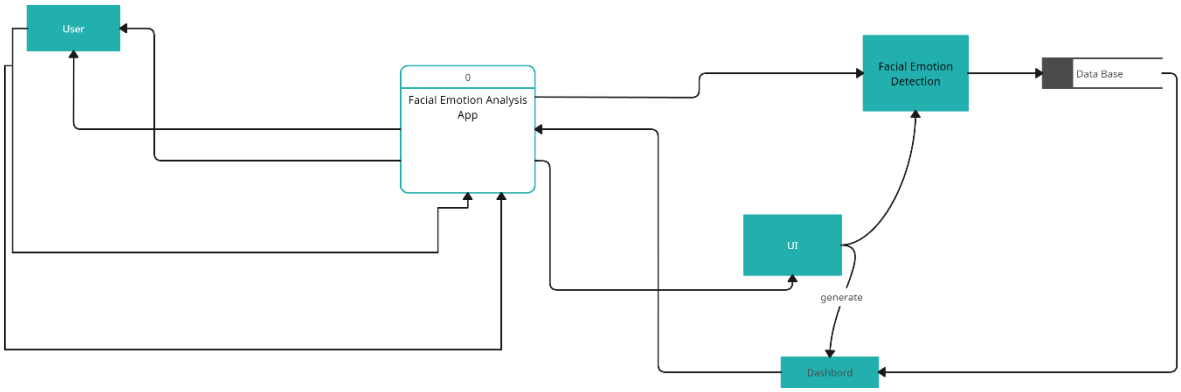

Figure 12 Data Flow Diagram

5.3.3 UML Use Case Diagram

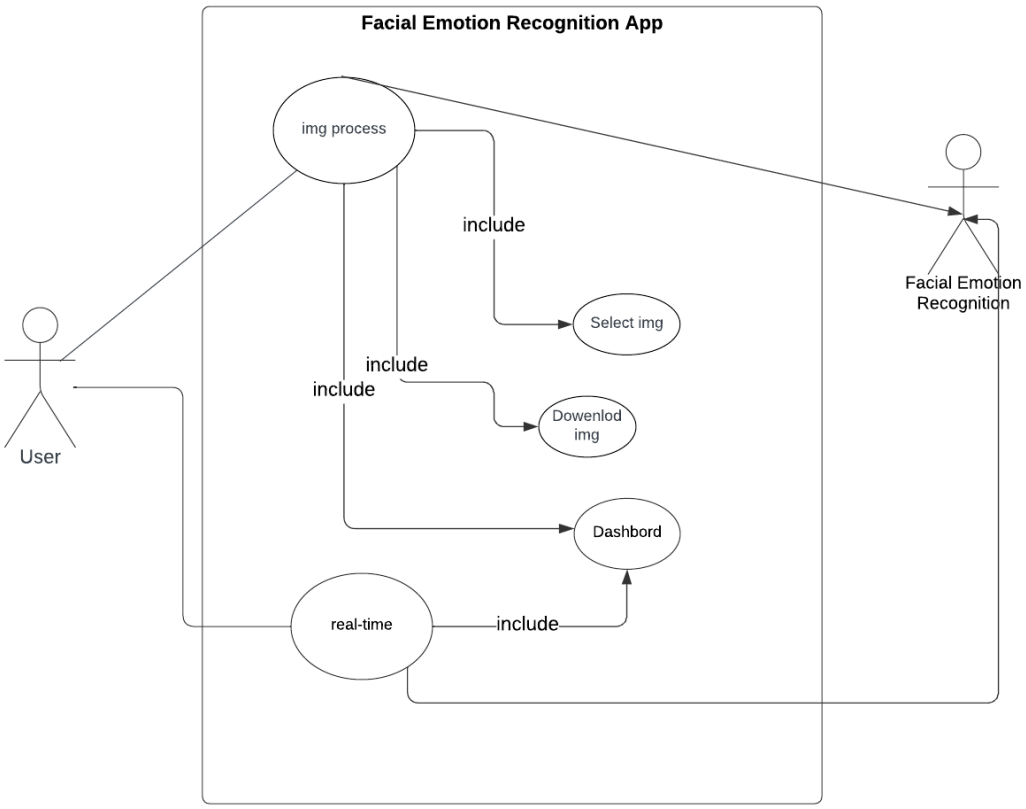

Figure 13 UML Use Case Diagram

5.3.4 activity Diagram

The activity diagram shows the sequence of activities that occur when a customer interacts with the FER system.

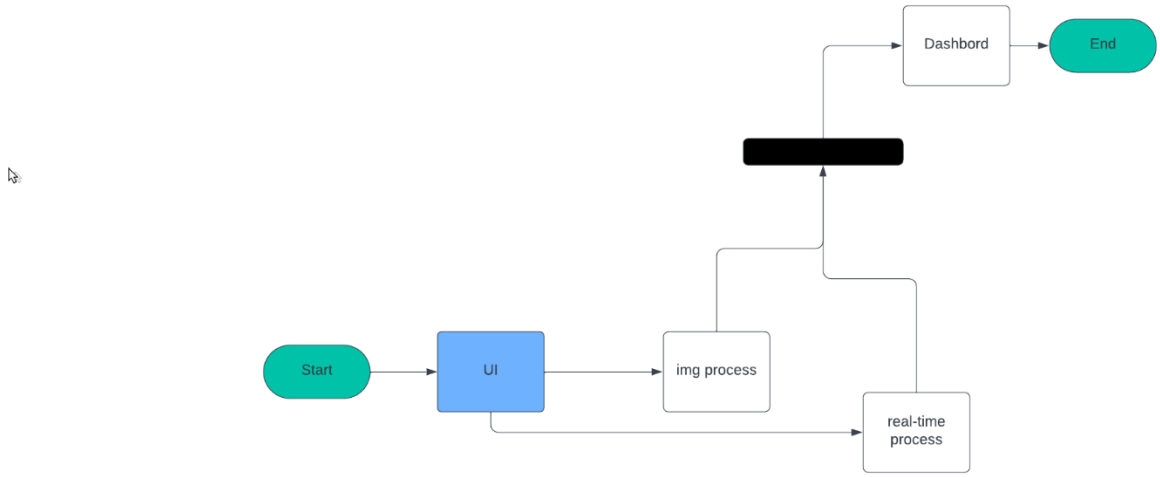

Figure 14 activity Diagram

5.3.5 Sequence diagram

The sequence diagram illustrates the interactions between the different classes in the FER system.

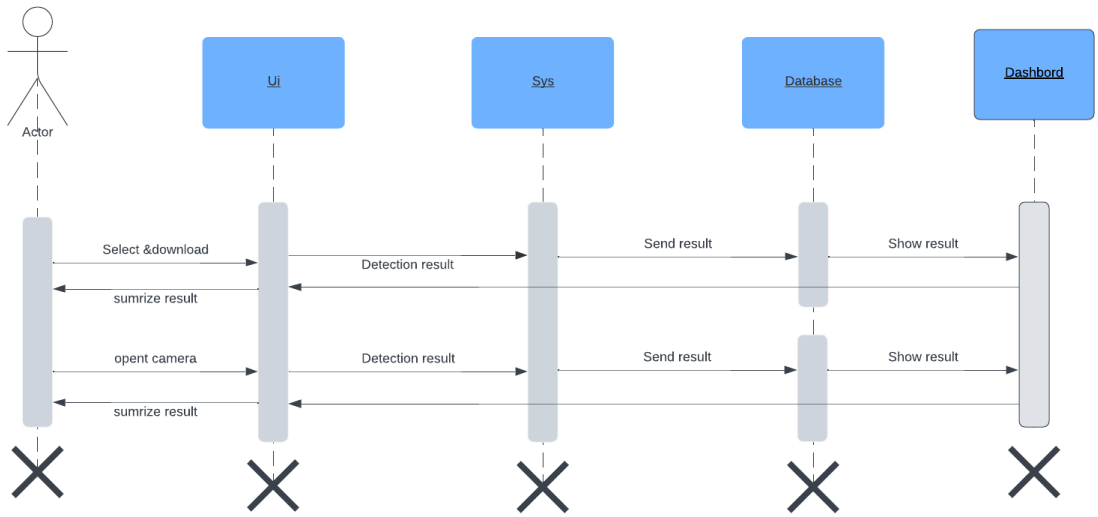

Figure 15 Sequence diagram

Chapter 6: Conclusions and Future Work

. Summary

This project aimed to develop a facial expression recognition (FER) system for customer satisfaction. The system utilizes machine learning algorithms to analyse facial expressions and identify customer emotions in real time. This information can be used by customer service representatives to better understand the needs of customers and provide better service.

System Overview

The FER system consists of three main components:

Face detection: This component detects faces in images and videos.

Facial feature extraction: This component extracts facial features from detected faces.

Emotion classification: This component classifies facial expressions into emotions.

The system also includes a performance monitoring component that tracks the accuracy of emotion recognition and provides feedback to customer service representatives.

System Implementation

The FER system was implemented using a combination of open-source and proprietary software libraries. The system was tested on a dataset of images of people expressing different emotions. The results showed that the system was able to accurately classify emotions with an average accuracy of -----%.

Conclusion

The FER system has the potential to improve customer satisfaction by providing customer service representatives with real-time feedback about customer emotions. This information can be used to provide more personalized and effective service.

Future Work

Future work on this project could include:

- Improving the accuracy of emotion recognition.
- Expanding the system to support a wider range of emotions.
- Integrating the system with other customer service applications.

Overall Significance and Contribution

This project has the potential to make a significant contribution to the field of customer service by providing a new tool for understanding customer emotions. This information can be used to improve customer satisfaction and loyalty.

Strengths and Limitations

strengths

The FER system has several strengths, including:

Accuracy: The system can accurately classify emotions with an average accuracy of -------%.

Real-time feedback: The system provides real-time feedback to customer service representatives.

limitations

The FER system also has some limitations, including:

Accuracy: The system is not always accurate, as it can be difficult to classify emotions based on facial expressions alone.

Scope: The system is currently limited to classifying a small number of emotions.

Integration: The system has not yet been integrated with other customer service applications.

Discussion of Potential Applications

The FER system has the potential to be used in a variety of customer service applications, including:

Call centres: The system can be used to provide real-time feedback to call centre representatives about customer emotions.

Retail stores: The system can be used to monitor customer emotions in retail stores and identify customers who may need assistance.

Analysis of Possible Future Research Directions

Future research on this project could focus on the following areas:

- Developing new methods for emotion recognition.
- Improving the accuracy of emotion recognition in real-time settings.
- Developing methods for detecting emotions from other modalities, such as speech and text.
- Studying the impact of customer emotions on customer satisfaction and loyalty.

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APPENDICES

The appendices to your report provide supporting evidence of the quality and quantity of the work you have done. All appendices in the project must be presented together after the concluding chapter. Page numbering for appendices continues on from the body of the project, in sequence.

This chapter can include one or more appendices. Each appendix should talk about one single subject that is discussed lightly within the report chapters.

Appendices must be limited to supporting material genuinely supplementary to the main argument of the project. Appendices must only include material that is referred to in the body of the project.

The following are appropriate for inclusion in the appendices:

- additional details of methodology and/or data
- diagrams of specialized equipment developed
- copies of questionnaires or surveys used in the research
- Source Code (including all scripts, makefiles, XML, HTML, etc.)
- If possible, include a compiled executable with running instructions
- Reading reports
- Interim Progress Reports
- Project Plan Document
- User Requirement Document (Development Project) or Theoretical Development (Research Project)
- Product/System Design Document
- Software High-Level Design
- Detailed Software Designs (for example, if using Java, include the Javadoc output files here)
- Software Test Documents (tests & test results)
- Software Evaluation Results
- Any additional outputs that you created during the project

Without this supporting evidence it is possible that the examiner will take the view that you have not done everything you claim to have done. However, the appendices are only there to back up the claims made in your report. Examiners can only be expected to look at those parts of the appendices you draw their attention to in the main body of the report. They are not obliged to read the appendices in detail, though they may do so. If

you think it is important to draw the markers' attention to a particular document, or a part of a document, tell them where to find (don't just say "the code for this is in appendix 3", give a page number, and/or other information that makes it clear how to find it). All program code written by you must be presented in the appendices. Do not include code that is machine generated, or that comes from a different author, unless it is necessary in order for the reader to understand the work you have done. If you do include code that you did not write yourself, it is your responsibility to make clear which parts of the program are your own and which parts are not. If you present automatically generated code, or the code of another programmer, as if it were your own, you may be accused of plagiarism. Do not include copies of any web pages that you have referred to, unless it is absolutely necessary for the reader to see them in order to make your point: just put the citation details in your bibliography. Samples of the work that is presented in the appendices may be included in the body of your report in order to illuminate a particular point or for discussion purposes.

1. Appendix A: Project Plan

2. Appendix B: User Requirements Specification

3. Appendix C: Project Design Specification

This is an outline report structure. Many projects will vary from this. Consult with your project supervisor to create the most appropriate structure for your report**.**